

Patrick PRELL

Robotics & Embedded Systems Engineer

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Hiring Manager
Company Name
Company Address

Dear Hiring Manager,

I am writing to express my strong interest in the Robotics Engineer position within Anduril's Maritime team. Over the past eight years, I have built and deployed robust embedded systems for mission-critical applications. I am eager to apply this experience to one of robotics' toughest challenges: ensuring autonomous surface and subsea vehicles operate with absolute reliability in harsh, unpredictable environments. My background spans complex RF hardware, safety-critical medical firmware, and field-tested UAV autonomy.

Signal Processing and Operational Efficiency (DroneShield): I own the antenna calibration pipeline for an ultra-wideband four-antenna direction-finding system. By resolving noisy multi-channel front-end interference, I ensured repeatable bearing accuracy across production units. Furthermore, by coordinating across production, hardware, and test engineering teams, I redesigned the calibration workflow, reducing execution time from twelve hours to just forty minutes per unit.

Safety-Critical Firmware and Rigor (ResusRight): I developed firmware for a clinical monitor compliant with IEC 62304 and ISO 13485 standards. Operating in an environment where failure is not an option gave me a deep, foundational discipline in safety architectures, field reliability, and rigorous validation qualities essential for autonomous maritime systems.

State Estimation, Computer Vision and Field Ops (CORDEL): Working with ArduPilot and PX4 across a survey UAV fleet, I handled everything from airframe bring-up to field operations. I designed an Extended Kalman Filter (EKF) fusing IMU and RTK-GNSS data for high-precision pose and position estimation, enabling accurate LiDAR point-cloud generation. Additionally, I built a C++/OpenCV pipeline for optical flow pose/velocity estimation and spent a year leading an autonomous project in New York extracting 3D geometry from moving train camera feeds.

End-to-End Autonomy and ROS 2: In my personal engineering work, I built a ROS 2 / PX4 SITL inspection stack for a VTOL platform. I use this system in the field to monitor stock and water levels on my family farm, giving me hands-on experience deploying ROS 2 nodes to practical, outdoor operations.

Anduril's focus on rapid iteration and software-defined defense hardware deeply aligns with how I build systems. I welcome the opportunity to discuss how my state estimation, embedded firmware, and field testing experience can contribute to the Maritime team.

Kind regards,
Patrick Prell